

ME350 FA 2025 Section 002

Lab #4

Monday 12:30 PM - 2:30 PM

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Agenda

- Reminders
- Project Schedule
- Manufacturing Review
- Lab 3 Recap
- Team Discussion/Work Time



Reminders

- Manufacturing Review is **Today in lab**
- **Design Report #1 due:**
 - Friday (9/26) - Monday Labs
- Be sure to reference source material to write Design Report (lecture slides, tutorials, etc.). Lots of resources under Project Resources on Canvas
- Project kits will be available for each team in the X50 Assembly Room
 - Cabinets under tables/ on sides of room
 - More materials will be given later in the semester
 - Other materials can be found in top 4 rows of grey cabinet



Project Schedule

High Level Overview

- By Friday/Thursday/Monday Sept. 15/ Sept. 19 Complete
Gate Review 1 with GSI/IA
- Friday/Thursday/Monday Sept. 15/ Sept. 3
Manufacturing Review in Lab 4
- Monday/Tuesday/Friday Sept. 22/Sept. 23 Sept. 26 DR 1
Due with First Draft of Drawings & Manufacturing Plan
completed
- Thursday Sept. 25 HW 3 (Motors) Due
- Friday/Thursday/Monday Oct. 2/Oct. 3/ Oct. 6 Gate
Review 2 in Lab 6
- Tuesday Sept. 30 MIDTERM EXAM
- Thursday/Friday/Monday Oct. 9/Oct.10/Oct. 15 Functional
Testing in Lab 7 (linkage parts + hardstops due)



Calendar

Sep 21	Sep 22	Sep 23	Sep 24	Sep 25	Sep 26
		Lecture 9 -Planetary Gears (DB)		Lecture 10 - Review for Midterm Exam (MU)	
				Homework 3 Due (Motors)	
	Lab 4 Manufacturing Review and Project Work Lab Assignment #3 Due - Individual ADAMS Graphs using Individual Solidworks Links			Lab 5 - Transmission Selection Tutorial	Lab 5 - Transmission Selection Tutorial
	DR 1 DUE TODAY AT MIDNIGHT Gate Review 1 Out of Lab-->	DR 1 DUE TODAY AT MIDNIGHT			DR 1 DUE TODAY AT MIDNIGHT
Sep 28	Sep 29	Sep 30	Oct 1	Oct 2	Oct 3
		Midterm Exam		Lecture 11: Selection of Motor and Transmission; Conflict Resolution (MU)	
					Homework 4 Released
	Lab 5 - Transmission Selection Tutorial			Lab 6 - Gate Review 2 in Lab (X50 Assembly Room, 1510 GGB) Lab Assignment #5 Due - Transmission Selection Workbook Bill of Materials (BOM) Transmission CAD	Lab 6 - Gate Review 2 in Lab (X50 Assembly Room, 1510 GGB) Lab Assignment #5 Due - Transmission Selection Workbook Bill of Materials (BOM) Transmission CAD
				Build & Assembly for Functional Testing -->	



Lab 3 Recap

Goal: Find the fastest mechanism with the same power limit

- θ : Angle sweep of input link from target 1 to 4 [deg]
- t : Total travel time [sec]
- α : Angular acceleration of input link [deg/sec²]

1. Assume 1-watt power limit

- a. What is P_{max} for your motor? What is your safety factor?
2. Measure θ from SW model - should be constant
 3. Decrease $t \rightarrow$ calculate $\alpha \rightarrow$ use in ADAMS motion function
 4. Simulate motion $\rightarrow$ plot power consumption $\rightarrow$ find $|P_{max}|$
 5. If $|P_{max}| < 1 \text{ W} \rightarrow$ return to step 3
 6. If $|P_{max}| \sim 1 \text{ W}$, t value is your shortest travel time



Manufacturing Review

- In lab 4.
- Goal: Discuss manufacturability with one other team + GSI/IA
 - Not a formal presentation: no need to dress up
 - **Attendance is mandatory** for all teams for all presentations
 - Every team gets 10 min to present (shorter is better) and 5 min for Q & A

Time Slot	Presenting Team (reviewing team)
12.45 pm - 1.00 pm	Team #1 (#7)
1.00 pm - 1.15 pm	Team #2 (#1)
1.15 pm - 1.30 pm	Team #3 (#2)
1.30 pm - 1.45 pm	Team #4 (#3)
1.45 pm - 2.00 pm	Team #5 (#4)
2.00 pm - 2.15 pm	Team #6 (#5)
2.15 pm - 2.30 pm	Team #7 (#6)



Appendix A: Assembly Manual

- Think IKEA
 - More pictures = easier for everyone to understand
 - Some supporting texts are helpful
 - Can tools/hands reach the necessary places for assembly?
- Break the assembly into smaller parts



Manufacturing Plan Approval Process



Manufacturing Approval Process

NEW DRAWINGS (NOT for drawing revisions)

- First, go to this [link](#)
 - You must be logged in to your umich account to submit
 - Don't use another person's account
- Fill out the form from the above link
 - Please use the drawing template found [here](#) on main Canvas
 - Project > Lab Materials > Manufacturing > Engineering Drawings
 - Each drawing must be **checked** by another team member before submission
- After you fill out this form, your GSI will be automatically emailed. Then either:
 - Drawing approved by GSI and is then sent to the machine shop for final approval
 - Your drawing is rejected
- Do not submit any drawings until you get feedback from your gate review



Rejected Drawings

If your drawing is rejected by your GSI, you will get an email that looks like this.

1. Click on the drawing to see your GSI's comments
2. Click on the link to submit your revised drawing

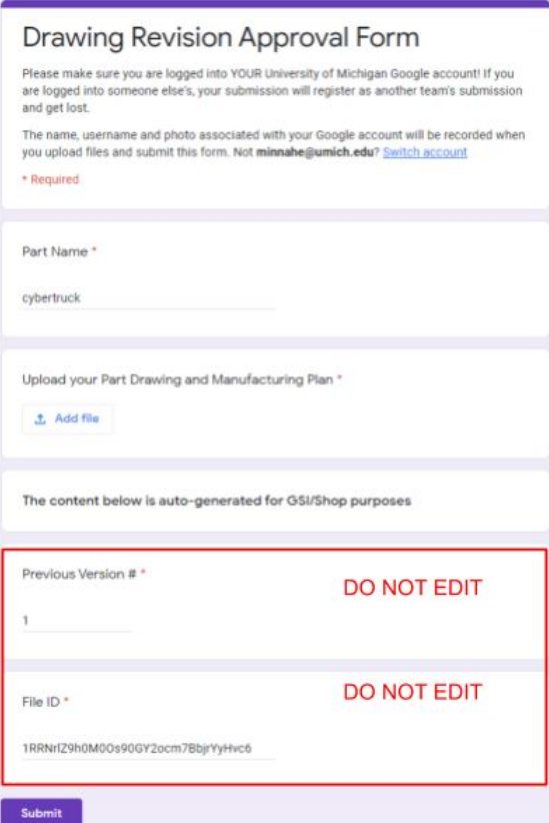


Submitting Revised Drawings

After clicking the link, the form will look like this and is **automatically pre-filled out for you!**

- All you need to do is upload your **revised** drawing and manufacturing plan
- **DO NOT EDIT the bottom two boxes**
 - This may break the auto-generated links, and your drawing may be lost!

You will go through the same process for drawings rejected by the machine shop, but your GSI **does not** need to re-approve the drawing.



The screenshot shows a web form titled "Drawing Revision Approval Form". It includes instructions for users to log into their University of Michigan Google account. The form has several sections: a "Part Name" field with the value "cybertruck", an "Upload your Part Drawing and Manufacturing Plan" section with an "Add file" button, and two sections at the bottom labeled "DO NOT EDIT". The first "DO NOT EDIT" section contains "Previous Version #" with the value "1". The second "DO NOT EDIT" section contains "File ID #" with a long alphanumeric string. A "Submit" button is at the bottom.

Drawing Revision Approval Form

Please make sure you are logged into YOUR University of Michigan Google account! If you are logged into someone else's, your submission will register as another team's submission and get lost.

The name, username and photo associated with your Google account will be recorded when you upload files and submit this form. Not minnahe@umich.edu? [Switch account](#)

* Required

Part Name *

cybertruck

Upload your Part Drawing and Manufacturing Plan *

[Add file](#)

The content below is auto-generated for GSI/Shop purposes

Previous Version # * **DO NOT EDIT**

1

File ID # **DO NOT EDIT**

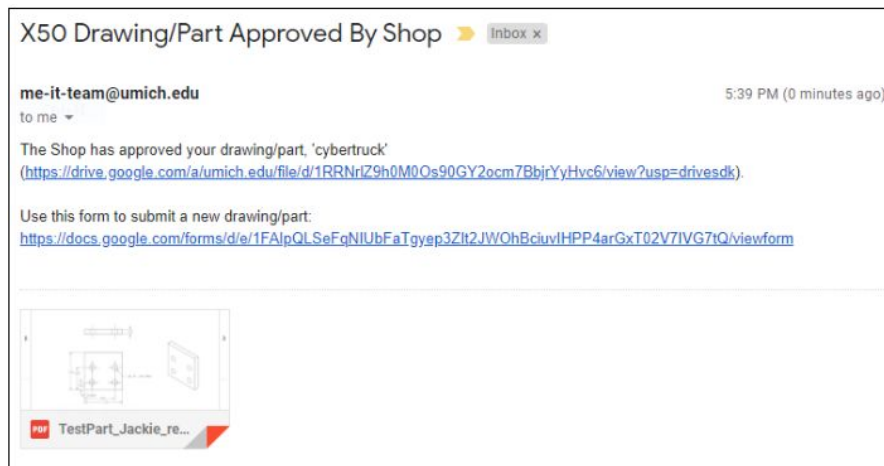
1RRNrI29h0M00s90GY2ocm7BbpYyHvc6

Submit

Approved Drawings

You will be email notified when your drawing is approved by both the GSI and the machine shop

- Make sure both signatures (your GSI and the machine shop) are in the title box



	NAME	DATE	ME 250 TEAM 92		
DRAWN	WAIERMA	11/5/19	TITLE: MOTOR BRACKET SPACER		
CHECKED	LIAMM	11/5/19			
GSI/IA APPR.	schmiedj	11/10			
SHOP APPR.	dwirkner	11/12/19			
INSPECTED					
COMMENTS:					
			SIZE A	DWG. NO. P92_006	REV 3
			SCALE: 4:1	QUANTITY: 1	SHEET 1 OF 3

Craftsmanship Grading

Judged by GSI/IA at the end of semester

- Were all the parts manufactured on time?
- Were there sharp edges or burrs?
- How was the surface finish?
- Were all the parts ultimately machined correctly?

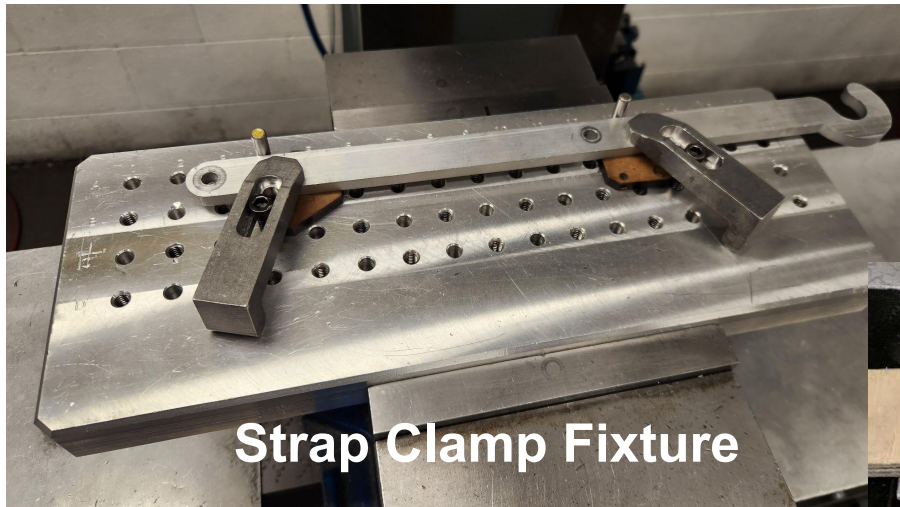


Long Link Manufacturing

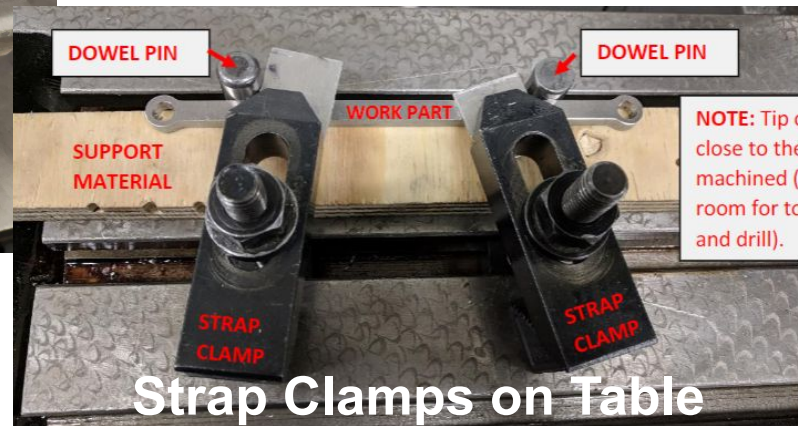
Strap Clamp (a.k.a Toe Clamp) Guide

For parts longer than the vise ($>6''$ long) or with no parallel holding features - an alternative fixturing method is required.

For parts that are between $6''$ and $13''$, check out a strap clamp fixture, which clamps in the vise on $1''$ parallels. For parts that do not fit on the fixture - use strap clamps on the mill table (remove vise).



Strap Clamp Fixture



Strap Clamps on Table

NOTE: Tip of straps contact close to the features to be machined (holes) but leave room for tool access (chuck and drill).

Long Link Manufacturing

- Use “Strap Clamp Guide” in Canvas for guidance on machining long parts
- Some ground plates may not fit in the vise!
 - Use this method for ground plates as well



Drawing/Manufacturing tips

- Think about how the part will be fixtured
 - Vise? Toe clamps? Are there clampable surfaces?
 - Is the part rigid enough to be clamped?
- Think about how indication will be performed
 - Is there a square corner for datuming?
- Think about how the 3D printer works
 - Is there material you can add or remove to eliminate the need for support material? Is a certain orientation preferred?
- Think about how to minimize opportunity for error by the machinist
 - Increasing diameter of endmills
 - Minimizing part removals from vise/fixture



Hard Stops, Spacers, and Motor

- Hard stops
 - They can be made from aluminum
 - **Think about adjustability of your hard stops...**
 - Can put rubber around them to soften the blow
 - You can mount these with a screw and a nut
- Mounting to Playing Field
 - The playing field has $\frac{3}{8}$ "-16 threads and $\frac{1}{4}$ " dowel pin holes
 - We have screws and dowels available
 - If you choose to use dowel pins:
 - You should have **at least two** screws connecting your mechanism to the playing field (for rigidity)
- Motor mounting
 - **Leave space to mount the motor with a bracket**
 - Motor CAD is available in the ME350 Hardware Files

Today

- Present Manufacturing Reviews
- Discuss group design with your team
- Work on DR1 (due Mon. 9/22 for Thursday Lab, due Tues. 9/23 for Friday Lab, due Fri. 9/26 for Monday Lab)
 - Review my DR1 feedback for submitted section - I will be looking for changes I recommended when I grade DR1!
 - Redo ADAMS for your team's best design if significant changes were made after GR1 or Manufacturing Review (i.e. link lengths changed, coupler design changed)
- Work on Drawings & Manufacturing Plans

Due next lab:

1. Nothing! Just work on...
 - a. DR1
 - b. Drawings
 - c. Manufacturing Plan

